Targeting the **Linux kernel** changes everything. Because the official Linux kernel source strictly rejects C++ (maintaining a pure C and Assembly codebase), you are likely either developing a custom out-of-tree kernel module, working on an embedded system with a specialized toolchain, or implementing a runtime layer.

When writing C++ for Linux kernel space, you must align your object models with the Linux kernel's internal memory design patterns.

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1. Match Your Object Sizes to Slab Buckets (kmalloc)

When you call kmalloc(size, flags), Linux doesn't allocate the exact byte size you request. Instead, it rounds up your request to the nearest **fixed-size power-of-two bucket** managed by the kmalloc-X slab caches (e.g., kmalloc-32, kmalloc-64, kmalloc-128, kmalloc-256).

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- **The Trap:** If your C++ class size is **65 bytes**, Linux will place it in the kmalloc-128 pool. This creates **63 bytes of internal fragmentation** per object.
- **The Fix:** Keep a close eye on your object layout. Use alignment attributes or organize data fields to keep your class sizes exactly at or slightly under a power of two.
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2. The Linux-Safe RAII Pattern (No Exceptions)

Because standard C++ smart pointers rely on std::bad_alloc exceptions when memory runs out, you must implement a lightweight, exception-free smart pointer using the **nothrow** pattern.

Here is how to implement a kernel-safe unique pointer for an object allocated via kmalloc:

```
#include <linux/slab.h>
#include <linux/types.h>

// Exception-free custom deleter for Linux kernel objects
template <typename T>
struct KernelDeleter {
    void operator()(T* ptr) const noexcept {
        if (ptr) {
            ptr->~T(); // Explicitly call the destructor
            kfree(ptr);
        }
    }
};
```

```cpp
// Lightweight unique pointer
template <typename T>
class KernelUniquePtr {
private:
    T* m_ptr = nullptr;

public:
    explicit KernelUniquePtr(T* ptr) noexcept : m_ptr(ptr) {}
    ~KernelUniquePtr() noexcept { Release(); }

    // Disallow copying to prevent double-frees
    KernelUniquePtr(const KernelUniquePtr&) = delete;
    KernelUniquePtr& operator=(const KernelUniquePtr&) = delete;

    // Allow moving
    KernelUniquePtr(KernelUniquePtr&& other) noexcept : m_ptr(other.m_ptr) { other.m_ptr
= nullptr; }

    void Release() noexcept {
        if (m_ptr) {
            KernelDeleter<T>{}(m_ptr);
            m_ptr = nullptr;
        }
    }

    T* Get() const noexcept { return m_ptr; }
    T& operator*() const noexcept { return *m_ptr; }
    T* operator->() const noexcept { return m_ptr; }
    explicit operator bool() const noexcept { return m_ptr != nullptr; }
};
```

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3. High-Frequency Allocations: Custom kmem_cache

If your driver creates and destroys thousands of instances of a specific C++ class, do not use kmalloc. Use the Linux kernel's built-in **Slab Cache system** (kmem_cache). It creates a dedicated memory pool tailored specifically to your object size, maximizing cache locality and zeroing out fragmentation.

Here is how to wrap a Linux Slab Cache inside a C++ static factory pattern:

```cpp
class NetworkPacket {
private:
    static struct kmem_cache* cp_cache; // Dedicated slab cache pointer
```

```
    // Private constructor ensures instantiation only goes through the custom allocator
    NetworkPacket() { /* Initialize properties */ }

public:
    ~NetworkPacket() { /* Release internal resources */ }

    // Initialize the slab cache when the module loads
    static int InitCache() {
        cp_cache = kmem_cache_create(
            "network_packet_cache",
            sizeof(NetworkPacket),
            0,                  // Alignment automatically handled
            SLAB_HWCACHE_ALIGN,    // Optimize for CPU L1/L2 cache lines
            nullptr               // Optional constructor callback
        );
        return cp_cache ? 0 : -ENOMEM;
    }

    // Destroy the cache when the module unloads
    static void DestroyCache() {
        if (cp_cache) kmem_cache_destroy(cp_cache);
    }

    // Factory method acting as a kernel-safe 'new'
    static KernelUniquePtr<NetworkPacket> Create(gfp_t flags) noexcept {
        if (!cp_cache) return KernelUniquePtr<NetworkPacket>(nullptr);

        void* raw_mem = kmem_cache_alloc(cp_cache, flags);
        if (!raw_mem) return KernelUniquePtr<NetworkPacket>(nullptr);

        // Placement new to invoke constructor safely without dynamic allocation
        NetworkPacket* obj = ::new (raw_mem) NetworkPacket();
        return KernelUniquePtr<NetworkPacket>(obj);
    }
};

// Define the static member
struct kmem_cache* NetworkPacket::cp_cache = nullptr;
```

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4. Memory Allocations vs. Execution Contexts

In Linux kernel development, your choice of allocation flags must perfectly match the context of the executing CPU thread. Choosing incorrectly will lock up or crash the OS.

Context	Allocation Flag	Behavior if Memory is Full	Can Sleep?	Safe for...
Process Context	GFP_KERNEL	Suspends thread until pages are freed.	**Yes**	IO tasks, Syscalls, Module Init
Atomic / Interrupt	GFP_ATOMIC	Instantly fails if emergency pool is dry.	**No**	ISRs, Tasklets, Inside Spinlocks
Hardware DMA	GFP_DMA	Allocates within the first 16MB zone of RAM.	**Yes**	Legacy hardware buffers

To take this to the next level, let me know:

- Are you writing an **out-of-tree character device driver**, a **network filter**, or something else?
- Do you need help configuring your **Makefile/Kbuild system** to link the C++ objects properly into the C-based kernel build environment?